IDENTIFICATION OF VALID EMAIL ADDRESSES USING LEX

Aim

To write and implement a LEX program to identify and print all valid-looking email addresses from a given text file.

Algorithm

1. Start the program.
2. Create a text file named input.txt containing text and email addresses.
3. Define a regular expression to recognize email address patterns.
4. Open the input file using the fopen() function.
5. Assign the opened file to yyin.
6. Call yylex() to scan the input file.
7. Print every matched email address using yytext.
8. Close the input file.
9. Stop the program.

Pseudocode

```
BEGIN
  OPEN input.txt for reading

  IF file cannot be opened
    PRINT error message
    STOP
  END IF

  SET yyin to input file
  SCAN input using yylex()

  FOR each match of the email regular expression
```

```
    PRINT the matched text (yytext)
  END FOR

  CLOSE input file
END
```

Program

Save the following program as email.l.

```
%{
#include <stdio.h>
%}

%%
[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,} {
  printf("Valid email: %s\n", yytext);
}
.|\n  ;
%%

int yywrap(void)
{
  return 1;
}

int main(void)
{
  FILE *fp = fopen("input.txt", "r");

  if (fp == NULL)
```

```
    {
        printf("Error opening input.txt\n");
        return 1;
    }

    yyin = fp;
    yylex();

    fclose(fp);
    return 0;
}
```

Sample Input

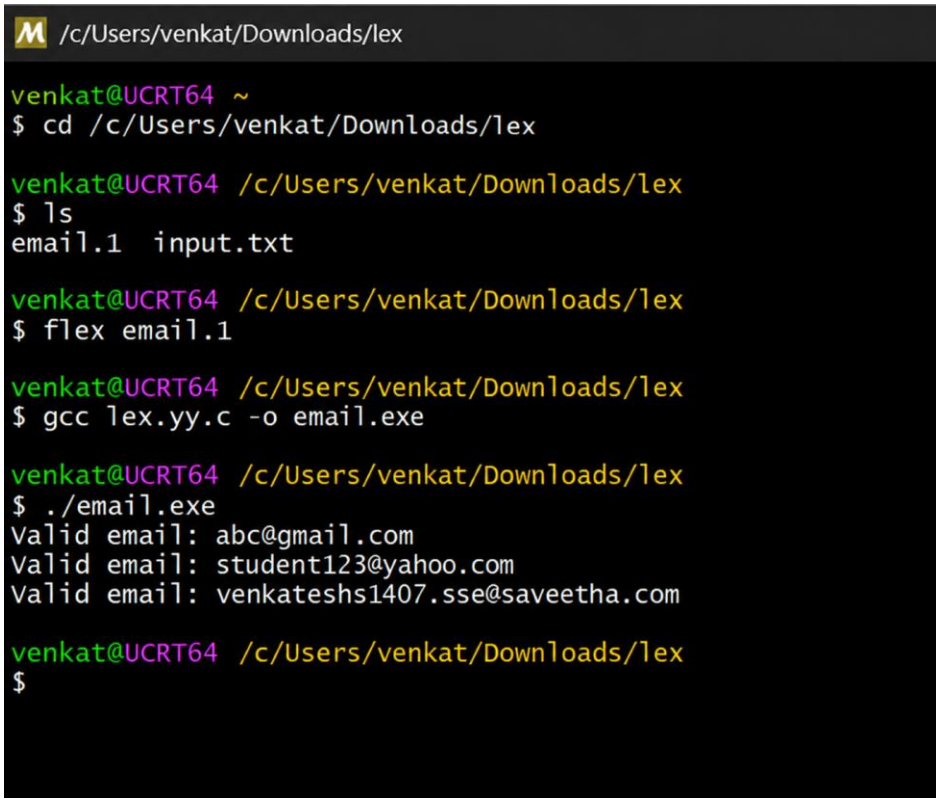
Contact: abc@gmail.com

Mail: student123@yahoo.com

Invalid: students@gmail

Another: venkateshs1407.sse@saveetha.com

Output



Result

Thus, the LEX program to identify and print valid-looking email addresses from the given text file was implemented successfully.